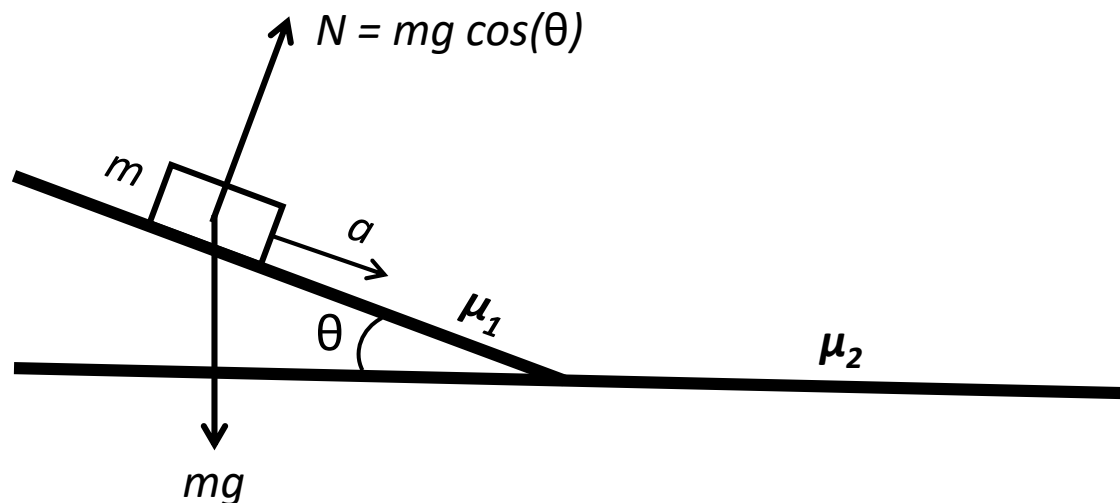


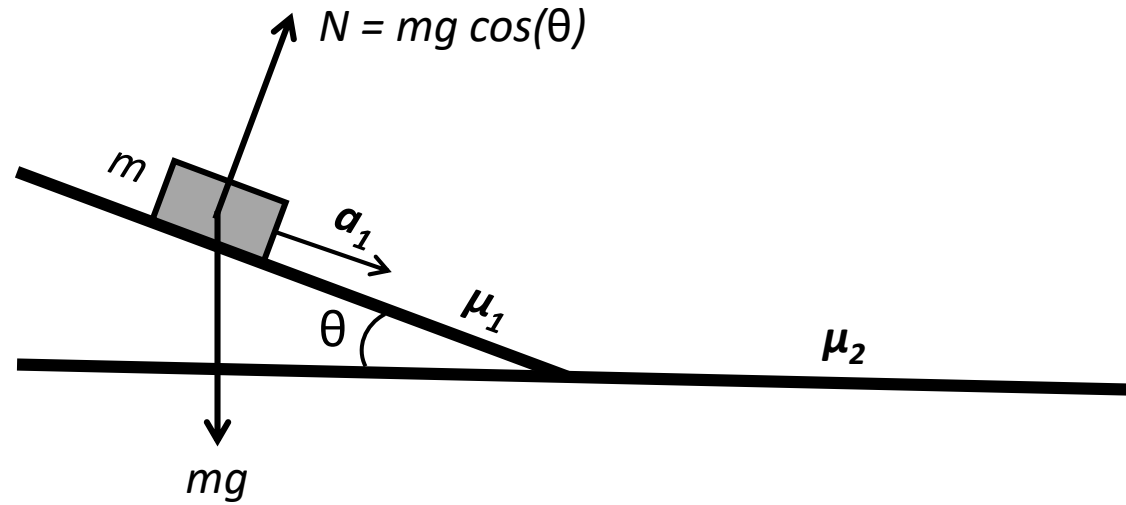
1.

Object Sliding Down an Inclined Plane

An object of mass $m = 4.0$ kg, starting from rest, slides down an inclined plane of length $l = 3.0$ m. The plane is inclined by an angle of $\theta = 30^\circ$ to the ground. The coefficient of kinetic friction $\mu_k = 0.2$. At the bottom of the plane, the mass slides along a rough surface with a coefficient of kinetic friction $\mu_k = 0.3$ until it comes to rest. The goal of this problem is to find out how far the object slides along the rough surface.



1.



Solution

$$m \cdot a_1 = m \cdot g \cdot \sin(\theta) - \mu_1 \cdot m \cdot g \cdot \cos(\theta)$$

$$a_1 = g \cdot (\sin(\theta) - \mu_1 \cdot \cos(\theta))$$

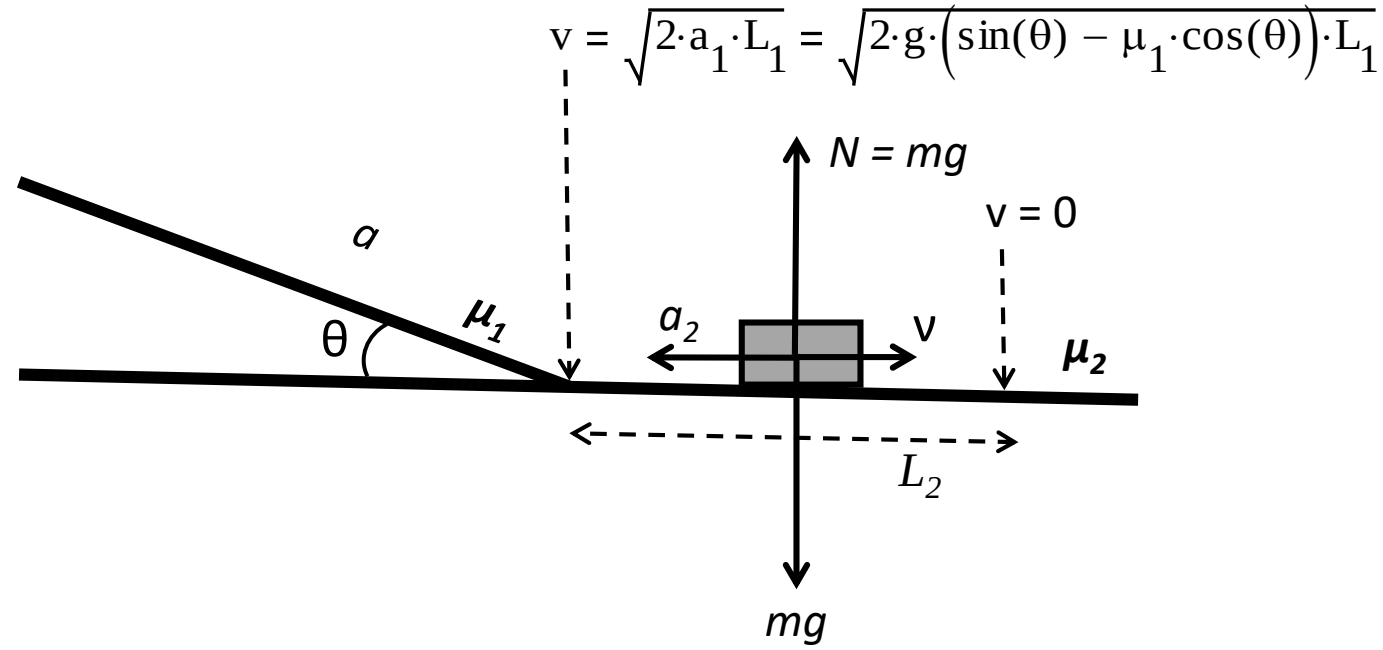
$$\begin{cases} L_1 = \frac{a_1 \cdot t^2}{2} \\ t = \frac{v}{a_1} \end{cases}$$



$$v = \sqrt{2 \cdot a_1 \cdot L_1} = \sqrt{2 \cdot g \cdot (\sin(\theta) - \mu_1 \cdot \cos(\theta)) \cdot L_1}$$

1.

Solution

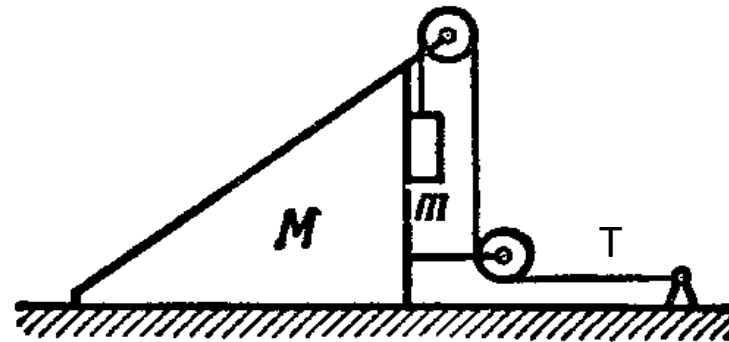


$$m \cdot a_2 = \mu_2 \cdot N = \mu_2 \cdot m \cdot g$$

$$a_2 = g \cdot \mu_2$$

$$L_2 = \frac{1}{2} \cdot \frac{v^2}{a_2} = \frac{1}{\mu_2} \cdot L_1 \cdot (\sin(\theta) - \mu_1 \cdot \cos(\theta))$$

2. The friction coefficient between the wedge (of mass M) and the bar (of mass m) is K . The masses of the thread and the block are negligible. Find the bar acceleration.



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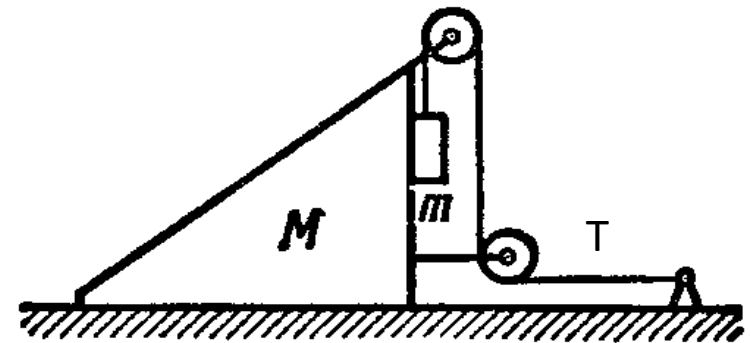
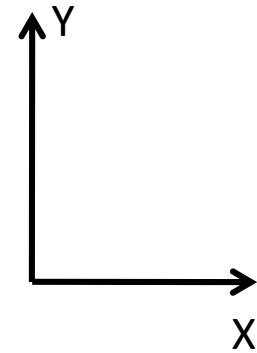
Solution

$$\left[\begin{array}{ll} a_x(m + M) = T & \text{/forces along X axis} \\ a_y m = mg - kN - T & \text{/forces along Y axis} \\ N = a_x m & \text{/horizontal normal force} \\ a_x = a_y = a \end{array} \right.$$

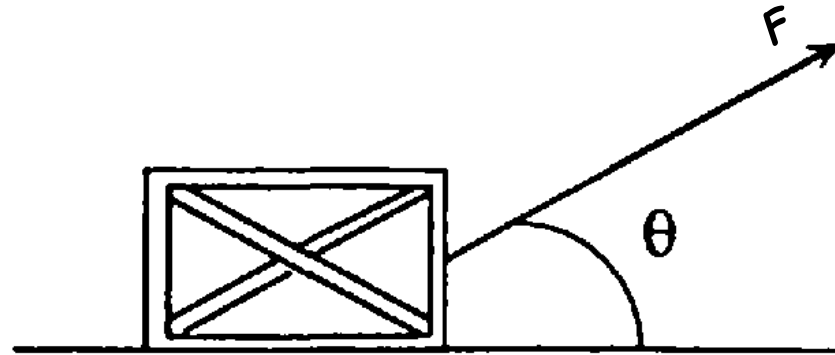
$$a(m + M) = mg - k(am) - am \quad \text{/master equation}$$

$$a = \frac{mg}{M + m(2 + k)}$$

$$\text{Answer: } a_{tot} = \sqrt{a_x^2 + a_y^2} = \frac{\sqrt{2} mg}{M + m(2 + k)}$$



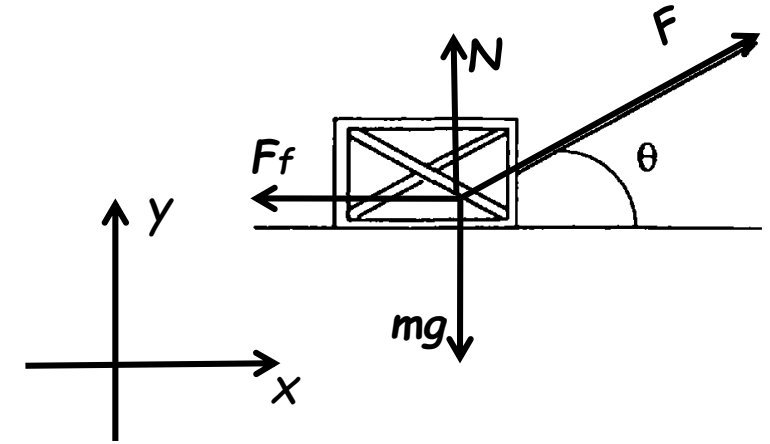
3. A packing crate of unknown mass is pulled across the floor at constant velocity by a cable attached to the crate. The angle θ_0 corresponds to the minimum force to move the crate. What is the friction coefficient between the crate and the floor?



Solution

3. A packing crate of unknown mass is pulled across the floor at constant velocity by a cable attached to the crate. The angle θ_0 corresponds to the minimum force to move the crate. What is the friction coefficient between the crate and the floor?

$$\begin{cases} F \cos(\theta) = \mu N & \text{/forces along X axis} \\ N = mg - F \sin(\theta) & \text{/forces along Y axis} \end{cases}$$



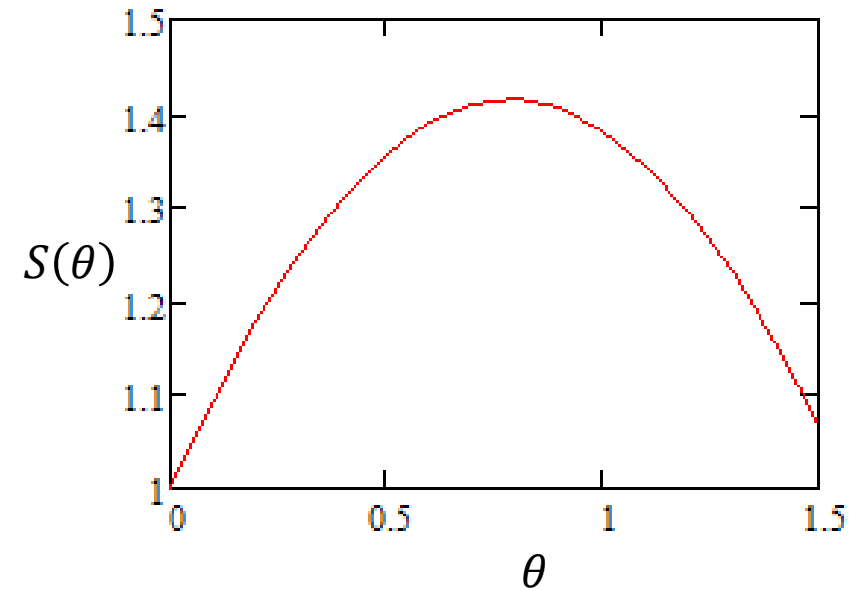
$$F(\cos(\theta) + \mu \sin(\theta)) = \mu mg \text{ /master equation}$$

$$F \cdot S(\theta) = \mu mg, \text{ where } S(\theta) = \cos(\theta) + \mu \sin(\theta)$$

$$\frac{dS(\theta)}{d\theta} = 0 \quad \text{/finding extrema of } S(\theta)$$

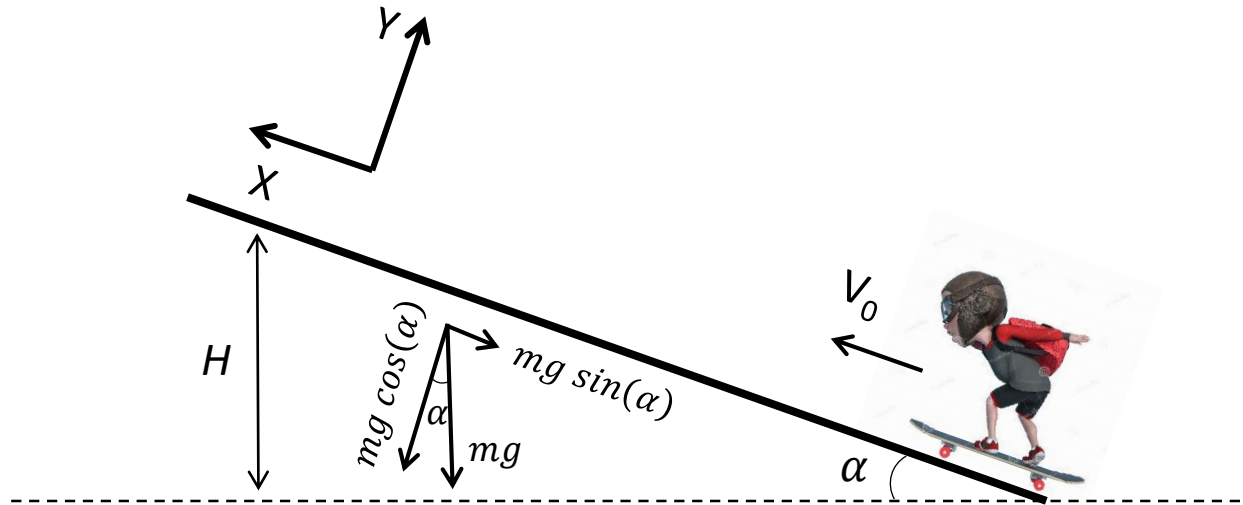
$$\mu \cos(\theta) - \sin(\theta) = 0$$

$$\text{Answer: } \mu = \tan(\theta)$$



4.

The man rides a skateboard. His initial velocity at the foot of hill is V_0 . What is the maximum height H he will reach, if the hill makes an angle α with the horizon? The friction coefficient of the skateboard is μ .



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Solution

$$N = mg \cos(\alpha) \quad \text{/normal reaction force}$$

$$-ma = -mg \sin(\alpha) - \mu N \quad \text{/dynamics along x-axis}$$

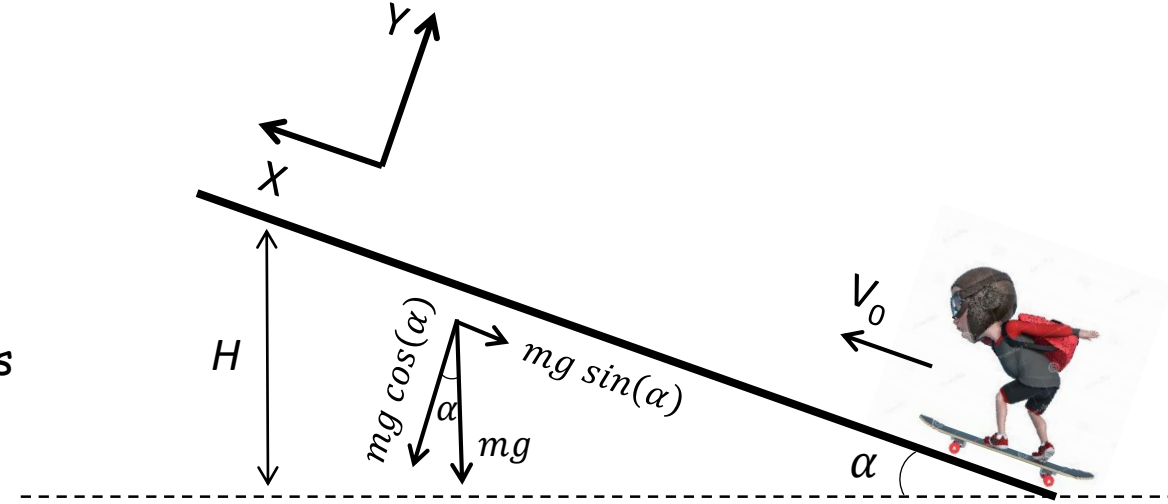
$$a = g (\sin(\alpha) + \mu \cos(\alpha)) \quad \text{/acceleration along x-axis}$$

$$t = \frac{v_0}{a} = \frac{v_0}{g (\sin(\alpha) + \mu \cos(\alpha))} \quad \text{/travel time}$$

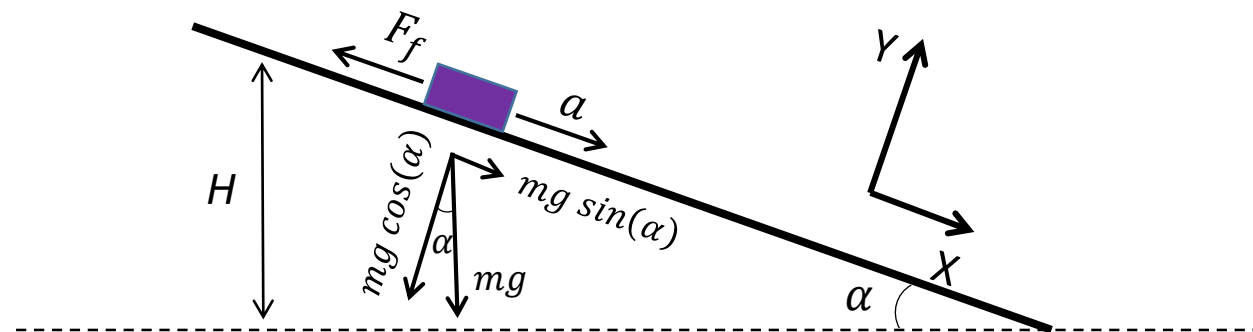
$$\begin{cases} S = \frac{H}{\sin(\alpha)} \\ S = \frac{at^2}{2} \end{cases} \quad \text{/expressions for total distance}$$

$$\frac{H}{\sin(\alpha)} = \frac{g (\sin(\alpha) + \mu \cos(\alpha)) v_0^2}{2g^2 (\sin(\alpha) + \mu \cos(\alpha))^2}$$

$$\text{Answer: } H = \frac{v_0^2}{2g(1 + \mu \cot(\alpha))}$$



5. The bar starts to slide down with a zero initial velocity along the plane with inclination of α with respect to horizon. The friction coefficient varies according to the expression $k(x) = f x$, where f is a positive constant, x is the distance traveled. Find (a) the total distance the bar traveled and (b) its maximum velocity.



5.

The bar starts to slide down with a zero initial velocity along the plane with inclination of α with respect to horizon. The friction coefficient varies according to the expression $k(x) = f x$, where f is a positive constant, x is the distance traveled. Find (a) the total distance the bar traveled and (b) its maximum velocity.

Solution (a)

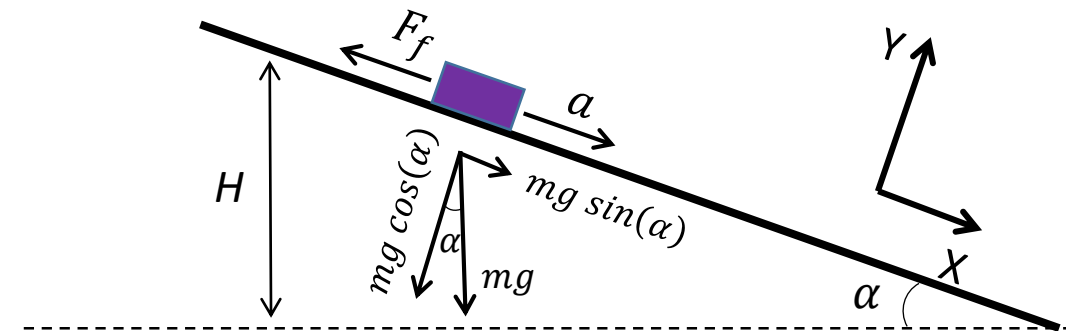
$$am = mg \sin(\alpha) - mg k(x) \cos(\alpha) \quad \text{/dynamics along x-axis}$$

$$v \frac{dv}{dx} = g(\sin(\alpha) - fx \cos(\alpha)) \quad \text{/used substitution } a = v \frac{dv}{dx}$$

$$v(x) = \sqrt{g(2\sin(\alpha)x - f\cos(\alpha)x^2)} \quad \text{/velocity as function of distance } x$$

$$v(D) = 0 \quad \text{/D is total distance} \quad \longrightarrow \quad (2\sin(\alpha) - f\cos(\alpha)D) = 0$$

Answer: final position of the bar is $D = \frac{2\sin(\alpha)}{f\cos(\alpha)}$



5. The bar starts to slide down with a zero initial velocity along the plane with inclination of α with respect to horizon. The friction coefficient varies according to the expression $k(x) = f x$, where f is a positive constant, x is the distance traveled. Find (a) the total distance the bar traveled and **(b)** its maximum velocity.

Solution (b)

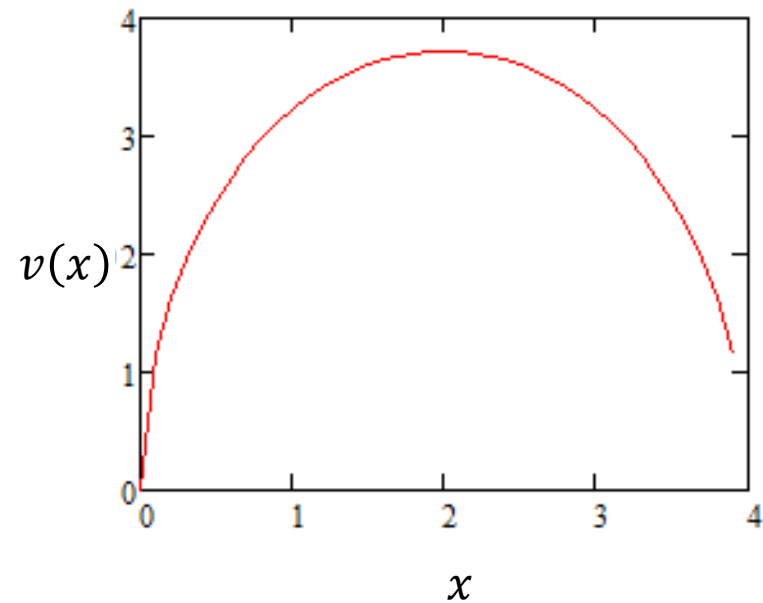
$$v(x) = \sqrt{g(2\sin(\alpha)x - f\cos(\alpha)x^2)} \quad \text{/velocity as function of distance } x$$

$$\frac{d}{dx'} (2\sin(\alpha)x' - f\cos(\alpha)x'^2) = 0 \quad \text{/finding extrema of } v(x) \text{ function}$$

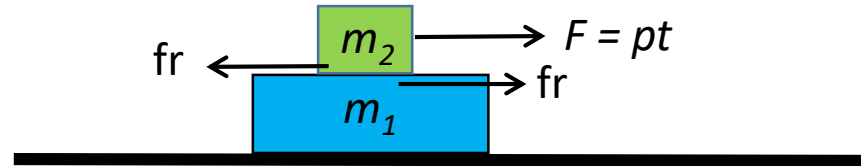
$$2\sin(\alpha) - 2f\cos(\alpha)x' = 0$$

$$x' = \frac{\sin(\alpha)}{f\cos(\alpha)} \quad \text{/velocity is maximum at } x', v_{max} = v(x')$$

Answer: maximum velocity is
$$v_{max} = \sin(\alpha) \sqrt{\frac{g}{f\cos(\alpha)}}$$



6. A plank of mass m_1 with a bar of mass m_2 placed on it lies on a smooth horizontal plane. A horizontal force growing with time t as $F = p t$ (p is constant) is applied to the bar. Find how the accelerations of the plank a_1 and of the bar a_2 depend on t , if the coefficient of friction between the plank and the bar is equal to k .



6.

A plank of mass m_1 with a bar of mass m_2 placed on it lies on a smooth horizontal plane. A horizontal force growing with time t as $F = p t$ (p is constant) is applied to the bar. Find how the accelerations of the plank a_1 and of the bar a_2 depend on t , if the coefficient of friction between the plank and the bar is equal to k .

First case $t \leq t'$, when $fr(t) = am_1$. Both masses have the same acceleration a and move as one unit. It means that the static friction force is less than its maximum value $fr(t) < fr_{max}$.

$fr_{max} = kN = km_2g$ /maximum value of the friction force

$fr(t) = F(t) - m_2a(t)$ /current value of the friction force

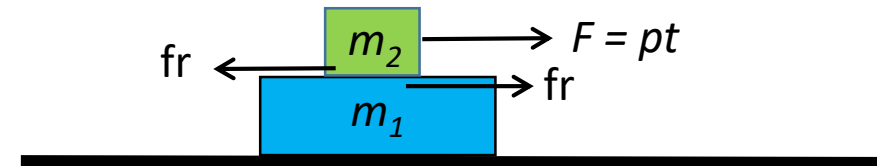
$$a(t) = \frac{F(t)}{(m_1 + m_2)} \quad \text{/acceleration of both masses}$$

Let's find t' :

$$\begin{cases} fr(t') = fr_{max} \\ fr(t') = pt' - m_2 \frac{pt'}{(m_1 + m_2)} \end{cases}$$

$$pt' - m_2 \frac{pt'}{(m_1 + m_2)} = km_2g$$

$$t' = \frac{km_2g(m_1 + m_2)}{pm_1}$$



Solution

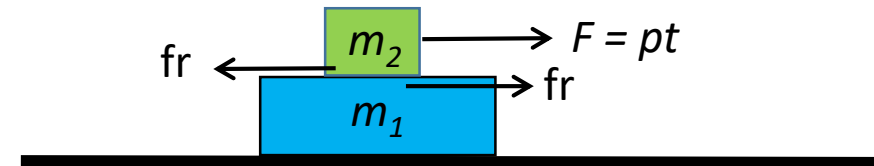
6.

A plank of mass m_1 with a bar of mass m_2 placed on it lies on a smooth horizontal plane. A horizontal force growing with time t as $F = p t$ (p is constant) is applied to the bar. Find how the accelerations of the plank a_1 and of the bar a_2 depend on t , if the coefficient of friction between the plank and the bar is equal to k .

Second case $t > t$. The mass m_1 has the constant acceleration: $fr_{max} = a_1 m_1$. The acceleration of the mass m_2 increases with time: $a_2(t) m_2 = F(t) - fr_{max}$

$$a_1 = \frac{fr_{max}}{m_1} = \frac{k m_2 g}{m_1}, \quad a_2(t) = \frac{pt - k m_2 g}{m_2}.$$

Answer:
$$\left\{ \begin{array}{l} t \leq \frac{k m_2 g (m_1 + m_2)}{p m_1} \\ a_1 = a_2 = \frac{pt'}{(m_1 + m_2)} \end{array} \right. ; \left\{ \begin{array}{l} t > \frac{k m_2 g (m_1 + m_2)}{p m_1} \\ a_1 = \frac{fr_{max}}{m_1} = \frac{k m_2 g}{m_1} \\ a_2(t) = \frac{pt - k m_2 g}{m_2}. \end{array} \right.$$



Solution